

2023 AMC 10B Problem 2 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 2. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 2

Problem:

Carlos went to a sports store to buy running shoes. Running shoes were on sale, with prices reduced by 20% on every pair of shoes. Carlos also knew that he had to pay a 7.5% sales tax on the discounted price. He had \$43. What is the original (before discount) price of the most expensive shoes he could afford to buy?

Answer Choices:

- A. \$46
- B. \$50
- C. \$48
- D. \$47
- E. \$49

Solution:

Let x represent the original price of the shoes in dollars. The discounted price is $0.8x$, and $1.075 \times 0.8x = 0.86x$ is the total amount Carlos has to pay. Therefore $0.86x$ must be less than or equal to 43, so

$$x \leq 43 \div 0.86 = 50$$

The original price of the most expensive shoes he could afford to buy is (B) \$50.

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